QUICK OPERATIONS & FEATURE MANUAL

# AI Dynamic 3D Camera App & Neo Board

Simple Step-by-Step Guide: App Usage, Features, Camera Ranges & Neo Board Execution

## 1. How to Use the AI Dynamic Cam App (Step-by-Step)

1. **Launch the App**: Open `index.html` in Google Chrome or Microsoft Edge, or launch the installed desktop PWA.
2. **Import 3D Model**: Drag & drop your `.glb` / `.gltf` 3D model into the landing drop zone (optionally attach background audio `.mp3` / `.wav` or camera JSON files). Click **Enter Editor Workspace**.
3. **Navigate 3D Viewport**:
   - **Rotate / Orbit**: Left-click and drag on the 3D viewport canvas.
   - **Pan Camera**: Right-click and drag (or `Shift` + Left-click and drag).
   - **Zoom In / Out**: Scroll mouse wheel up / down.
   - **Fit / Reset View**: Press `F` to fit model to screen; press `R` to reset camera view.
4. **Add Timeline Keyframes**:
   - Scrub the timeline playhead to any frame (0 to 1000).
   - Position the camera pose in the viewport and press `K` (or click **+ Keyframe**).
   - Press `Space` to Play / Pause timeline animation playback.
   - Press `Shift + K` to delete the keyframe at the current playhead frame.

## 2. Key Features Overview

### 🤖 AI Camera Director

Type natural prompts (e.g. "*Hero push in with 35mm lens*") or pick motion archetypes (*Hero Push*, *Wide Establish*, *Close Detail*, *Orbit Sweep*, *Spiraling Reveal*).

### 📷 Lens Focal Length (mm)

Adjust focal length from `1mm` to `120mm`. Quick presets: `24mm` (wide), `35mm` (natural), `50mm` (standard), `85mm` (detail portrait).

### 🎬 Handheld Shake & Audio Sync

Apply procedural shake profiles (*Continuous Handheld*, *Impact Drop*, *Erratic Rumbling*) synced with audio track waveform scrubbing.

### 💾 Project Export / Import

Export clean keyframe arrays for Three.js/Blender or download full project `.json` files containing all paths, hotspots, and settings.

## 3. Camera Ranges: Restricting Dynamic Camera Motion

> 🎯 **Core Purpose of Camera Ranges:**
> If you do **NOT** want dynamic camera motion active across the entire 3D model, you can create specific **Camera Ranges**. Dynamic camera motion will take effect **ONLY inside the added Camera Ranges** in Neo Board. Outside of these added range blocks, the camera stays steady in a fixed static pose. This allows Neo Board text banners, product specifications, and callouts to display cleanly without unwanted camera movement across the full model!

### How to Add & Configure Camera Ranges (Step-by-Step):

1. Identify the specific frame window where you WANT dynamic camera motion on your model (e.g. Frame 60 to Frame 180 for a 360° product sweep).
2. Open the **Camera Ranges** panel in the right drawer inspector.
3. Click **+ Add Range** to open the inline range editor form.
4. Enter your **Start Frame** (e.g. `60`) and **End Frame** (e.g. `180`).
5. Configure **Dynamic Tracking Mode**:
   - **Checked (`dynamic: true`)**: Dynamic camera motion takes effect **ONLY inside this range** in Neo Board.
   - **Unchecked (`dynamic: false`)**: Locks to a fixed static 3D pose, keeping the viewport steady outside dynamic zones.
6. Click **Save**. (You can also click directly on empty timeline track lane space to spawn a range block or toggle dynamic mode).

### How to Edit an Existing Camera Range:

- **Click Edit on Range Item**: In the Camera Ranges list, click the **Edit** button next to any saved range.
- **Modify Settings**: The form opens in **Edit Range** mode pre-populated with that range's Start Frame, End Frame, Dynamic Tracking checkbox, and Static pose values.
- **Save Changes**: Adjust any parameters and click **Save** to update that specific camera range in real time!

## 4. What Effect Happens in Neo Board?

**Neo Board** is the app's integrated presentation overlay HUD system ( `.neo-overlay` , `.neo-badge` , `.neo-title` , `.neo-desc` , `.neo-tags` ).

| Camera Range Mode | 3D Viewport Motion | Neo Board Overlay Effect |
|---|---|---|
| **Inside Added Camera Range** `(dynamic: true)` | Camera actively orbits, dollies, or sweeps along calculated 3D spline curves. | Displays active real-time motion badges (e.g., "*DYNAMIC SWEEP*") and updates spatial context tags dynamically as the camera sweeps. |
| **Outside Added Camera Range** `(dynamic: false)` | Camera motion is frozen; holds exact static orientation (Yaw, Pitch, Radius, Target XYZ, 0 Shake). | **Freezes viewport movement** so viewers can comfortably read title cards, specs, and text callouts without camera jitter or motion blur. |

> **Pro Tip:** Add Dynamic Camera Ranges only on segments where you need active 3D camera sweeps, and leave static callout segments un-ranged so text stays steady!

## 5. Keyboard Shortcuts Quick Reference

| Key | Action |
| --- | --- |
| `Space` | Play / Pause timeline animation playback |
| `K` | Insert keyframe at active playhead frame |
| `Shift + K` | Delete keyframe at active playhead frame |
| `Left` / `Right` | Step playhead 1 frame backward / forward |
| `P` | Toggle Preview Mode (hides drawer UI for clean presentation) |
| `F` | Fit 3D model geometry to viewport center |
| `R` | Reset camera view to default orientation |